

# NEET Chemistry — Chemical Bonding

Ultra Premium Practice Sheet (20 Questions)

**Q1. Which has maximum lattice energy?**

- A. NaCl
- B. MgO
- C. KCl
- D. CaO

**Q2. Hybridization of  $\text{SF}_6$  is:**

- A.  $\text{sp}^3$
- B.  $\text{sp}^3\text{d}$
- C.  $\text{sp}^3\text{d}^2$
- D.  $\text{sp}^2$

**Q3. Bond order of  $\text{O}_2^{2+}$  is:**

- A. 1.5
- B. 2
- C. 2.5
- D. 3

**Q4. Which is most polar?**

- A.  $\text{CO}_2$
- B.  $\text{H}_2\text{O}$
- C.  $\text{BF}_3$
- D.  $\text{CH}_4$

**Q5. Shape of  $\text{XeF}_4$  is:**

- A. Linear
- B. Trigonal planar
- C. Tetrahedral
- D. Bent

**Q6. Assertion:  $\text{NH}_3$  is polar. Reason: It has lone pair on N.**

- A. Both correct explain
- B. Both correct not explain
- C. A correct R wrong
- D. Both wrong

**Q7. Which has highest bond dissociation energy?**

- A.  $\text{F}_2$
- B.  $\text{Cl}_2$
- C.  $\text{Br}_2$
- D.  $\text{I}_2$

**Q8. Geometry of  $\text{CH}_4$  is:**

- A. Tetrahedral
- B. Planar
- C. Linear
- D. Bent

**Q9. Which violates octet rule?**

- A.  $\text{BF}_3$
- B.  $\text{NH}_3$
- C.  $\text{H}_2\text{O}$
- D.  $\text{CH}_4$

**Q10. Dipole moment is zero in:**

- A.  $\text{NH}_3$
- B.  $\text{H}_2\text{O}$
- C.  $\text{CO}_2$
- D.  $\text{SO}_2$

**Q11. Which has  $\text{sp}^2$  hybridization?**

- A.  $\text{CO}_2$
- B.  $\text{BF}_3$
- C.  $\text{NH}_3$
- D.  $\text{CH}_4$

**Q12. Bond angle in  $\text{H}_2\text{O}$  is:**

- A.  $104.5^\circ$
- B.  $109.5^\circ$
- C.  $120^\circ$
- D.  $180^\circ$

**Q13. Strongest intermolecular force:**

- A. H-bond

- B. Dipole
- C. Dispersion
- D. Ionic

**Q14. Which has resonance?**

- A.  $\text{O}_2$
- B.  $\text{NH}_3$
- C.  $\text{CH}_4$
- D.  $\text{H}_2$

**Q15. Bond order = 0.5 exists in:**

- A.  $\text{H}_2$
- B.  $\text{He}_2$
- C.  $\text{O}_2$
- D.  $\text{N}_2$

**Q16. Which has maximum dipole moment?**

- A. HF
- B. HCl
- C. HBr
- D. HI

**Q17. Hybridization of  $\text{CO}$  carbon:**

- A.  $sp$
- B.  $sp^2$
- C.  $sp^3$
- D. None

**Q18. Which has trigonal planar shape?**

- A.  $\text{BF}_3$
- B.  $\text{NH}_3$
- C.  $\text{H}_2\text{O}$
- D.  $\text{CH}_4$

**Q19. Lone pair repulsion order:**

- A.  $\text{LP-LP} > \text{LP-BP} > \text{BP-BP}$
- B.  $\text{BP-BP} > \text{LP-BP} > \text{LP-LP}$
- C.  $\text{LP-BP} > \text{BP-BP} > \text{LP-LP}$
- D. None

**Q20. Which has highest boiling point?**

- A.  $\text{NH}_3$  ☐
- B.  $\text{PH}_3$  ☐
- C.  $\text{AsH}_3$  ☐
- D.  $\text{SbH}_3$  ☐

## Answer Key

Q1: B  
Q2: B  
Q3: A  
Q4: B  
Q5: A  
Q6: A  
Q7: B  
Q8: A  
Q9: A  
Q10: C  
Q11: B  
Q12: A  
Q13: A  
Q14: A  
Q15: A  
Q16: A  
Q17: A  
Q18: A  
Q19: A  
Q20: D